

Topic 1: Basic Linux Commands

```
student@DESKTOP-UIMU3UD:~$ pwd
/home/student
```

```
student@DESKTOP-UIMU3UD:~$
```

```
student@DESKTOP-UIMU3UD:~$ history
 1 nano Hello.c
 2 pwd
 3 clear
 4 history
 5 whoami
 6 date
 7 cal
 8 sudo apt install ncal
 9 pwd
10 clear
11 pwd
12 clear
13 history
```

```
student@DESKTOP-UIMU3UD:~$ whoami
student
student@DESKTOP-UIMU3UD:~$ date
Sun Aug  9 08:27:32 UTC 2026
```

```
student@DESKTOP-UIMU3UD:~$ cal
      August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
student@DESKTOP-UIMU3UD:~$ hostname
DESKTOP-UIMU3UD
```

```
student@DESKTOP-UIMU3UD:~$ uname -a
Linux DESKTOP-UIMU3UD 4.4.0-19041-Microsoft #4355-Microsoft Thu Apr 12 17:37:00 PST 2024 x86_64 x86_64 x86_64 GNU/Linux
```

Topic 2: Directory Management

```

student@DESKTOP-UIMU3UD:~$ ls
Hello.c      exec_example.c  fork_exec2      fork_test.c     name.c          parent_child_pid  pid.c          zombie
Hello.c.save fork_exec        fork_exec2.c    hello           parent_child     parent_child_pid.c  wait_example    zombie.c
exec_example  fork_exec.c      fork_test       name            parent_child.c   pid                wait_example.c
student@DESKTOP-UIMU3UD:~$ ls -l
total 176
-rw-r--r-- 1 student student 76 Aug 8 15:19 Hello.c
-rw-r--r-- 1 student student 155 Aug 8 15:14 Hello.c.save
-rwxr-xr-x 1 student student 16008 Aug 9 06:07 exec_example
-rw-r--r-- 1 student student 159 Aug 9 06:07 exec_example.c
-rwxr-xr-x 1 student student 16088 Aug 9 05:23 fork_exec
-rw-r--r-- 1 student student 259 Aug 9 05:23 fork_exec.c
-rwxr-xr-x 1 student student 16088 Aug 9 06:10 fork_exec2
-rw-r--r-- 1 student student 264 Aug 9 06:09 fork_exec2.c
-rwxr-xr-x 1 student student 16008 Aug 9 05:17 fork_test
-rw-r--r-- 1 student student 101 Aug 9 05:17 fork_test.c
-rwxr-xr-x 1 student student 15960 Aug 8 15:20 hello
-rwxr-xr-x 1 student student 16064 Aug 8 15:26 name
-rw-r--r-- 1 student student 154 Aug 8 15:25 name.c
-rwxr-xr-x 1 student student 16008 Aug 9 05:21 parent_child
-rw-r--r-- 1 student student 226 Aug 9 05:20 parent_child.c
-rwxr-xr-x 1 student student 16104 Aug 9 06:02 parent_child_pid
-rw-r--r-- 1 student student 287 Aug 9 06:01 parent_child_pid.c
-rwxr-xr-x 1 student student 16000 Aug 8 15:28 pid
-rw-r--r-- 1 student student 108 Aug 8 15:28 pid.c
-rwxr-xr-x 1 student student 16048 Aug 9 06:04 wait_example
-rw-r--r-- 1 student student 273 Aug 9 06:04 wait_example.c
-rwxr-xr-x 1 student student 16088 Aug 9 05:26 zombie
-rw-r--r-- 1 student student 303 Aug 9 05:26 zombie.c
student@DESKTOP-UIMU3UD:~$ ls -a
.      .local      exec_example  fork_test     parent_child.c  wait_example.c
..     .motd shown exec_example.c fork_test.c     parent_child_pid  zombie
.bash_history .profile     fork_exec     hello         parent_child_pid.c  zombie.c
.bash_logout .sudo_as_admin_successful fork_exec.c    name          pid
.bashrc      Hello.c      fork_exec2    name.c        pid.c
.landscape   Hello.c.save fork_exec2.c   parent_child  wait_example
student@DESKTOP-UIMU3UD:~$ mkdir OSLab
student@DESKTOP-UIMU3UD:~$ cd OSLab
student@DESKTOP-UIMU3UD:~/OSLab$ cd ..
student@DESKTOP-UIMU3UD:~$ rmdir Test
rmdir: failed to remove 'Test': No such file or directory

```

```

student@DESKTOP-UIMU3UD:~$ find . -name "*.c"
./exec_example.c
./fork_exec.c
./fork_exec2.c
./fork_test.c
./Hello.c
./name.c
./parent_child.c
./parent_child_pid.c
./pid.c
./wait_example.c
./zombie.c

```

```

student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$

```

Topic 3: File Management

```

student@DESKTOP-UIMU3UD:~$ touch file1.txt
student@DESKTOP-UIMU3UD:~$ cat file1.txt
student@DESKTOP-UIMU3UD:~$ cp file1.txt file2.txt
student@DESKTOP-UIMU3UD:~$ mv file2.txt renamed.txt
student@DESKTOP-UIMU3UD:~$ rm renamed.txt
student@DESKTOP-UIMU3UD:~$ head Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}
student@DESKTOP-UIMU3UD:~$ tail Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}
student@DESKTOP-UIMU3UD:~$ more Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}

```

```

student@DESKTOP-UIMU3UD:~$ less Hello.c
student@DESKTOP-UIMU3UD:~$ wc Hello.c
  6  9 76 Hello.c
student@DESKTOP-UIMU3UD:~$ diff Hello.c name.c
4c4,7
<     printf("Hello Linux\n");
---
>     char name[30];
>     printf("Enter your name: ");
>     scanf("%s",name);
>     printf("Welcome %s\n",name);
student@DESKTOP-UIMU3UD:~$ sort file1.txt
student@DESKTOP-UIMU3UD:~$ uniq file1.txt
student@DESKTOP-UIMU3UD:~$ grep main Hello.c
int main()
student@DESKTOP-UIMU3UD:~$

```

Topic 4: File Editing

```

student@DESKTOP-UIMU3UD:~$ nano hello.c
student@DESKTOP-UIMU3UD:~$ vi test.c
student@DESKTOP-UIMU3UD:~$ nano hello.c
student@DESKTOP-UIMU3UD:~$ vi test.c
student@DESKTOP-UIMU3UD:~$ history | tail -n 5
 49  nano hello.c
 50  vi test.c
 51  nano hello.c
 52  vi test.c
 53  history | tail -n 5
student@DESKTOP-UIMU3UD:~$

```

Topic 5: C Program Compilation and Execution

```

student@DESKTOP-UIMU3UD:~$ echo -e '#include <stdio.h>\nint main() { printf("Hello World\\n"); return 0; }' > hello.c
student@DESKTOP-UIMU3UD:~$ gcc hello.c -o hello
student@DESKTOP-UIMU3UD:~$ ./hello
Hello World
student@DESKTOP-UIMU3UD:~$ which gcc
/usr/bin/gcc
student@DESKTOP-UIMU3UD:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz
student@DESKTOP-UIMU3UD:~$ man gcc
student@DESKTOP-UIMU3UD:~$

```

```

GCC(1)                                GNU                                GCC(1)
NAME
    gcc - GNU project C and C++ compiler

SYNOPSIS
    gcc [-c|-S|-E] [-std=standard]
        [-g] [-pg] [-Olevel]
        [-Wwarn...] [-Wpedantic]
        [-Idir...] [-Ldir...]
        [-Dmacro[=defn]...] [-Umacro]
        [-foption...] [-mmachine-option...]
        [-o outfile] [@file] infile...

    Only the most useful options are listed here; see below for the remainder.  g++ accepts mostly the same options as gcc.

DESCRIPTION
    When you invoke GCC, it normally does preprocessing, compilation, assembly and linking.  The "overall options" allow you to stop this process at an
    intermediate stage.  For example, the -c option says not to run the linker.  Then the output consists of object files output by the assembler.

    Other options are passed on to one or more stages of processing.  Some options control the preprocessor and others the compiler itself.  Yet other options
    control the assembler and linker; most of these are not documented here, since you rarely need to use any of them.

    Most of the command-line options that you can use with GCC are useful for C programs; when an option is only useful with another language (usually C++), the
    explanation says so explicitly.  If the description for a particular option does not mention a source language, you can use that option with all supported
    languages.

    The usual way to run GCC is to run the executable called gcc, or machine-gcc when cross-compiling, or machine-gcc-version to run a specific version of GCC.
    When you compile C++ programs, you should invoke GCC as g++ instead.

    The gcc program accepts options and file names as operands.  Many options have multi-letter names; therefore multiple single-letter options may not be
    grouped: -dv is very different from -d -v.

    You can mix options and other arguments.  For the most part, the order you use doesn't matter.  Order does matter when you use several options of the same
    kind; for example, if you specify -L more than once, the directories are searched in the order specified.  Also, the placement of the -l option is
    significant.

    Many options have long names starting with -f or with -W--for example, -fmove-loop-invariants, -Wformat and so on.  Most of these have both positive and
    negative forms; the negative form of -ffoo is -fno-foo.  This manual documents only one of these two forms, whichever one is not the default.

    Some options take one or more arguments typically separated either by a space or by the equals sign (=) from the option name.  Unless documented otherwise,
    an argument can be either numeric or a string.  Numeric arguments must typically be small unsigned decimal or hexadecimal integers.  Hexadecimal arguments
    must begin with the 0x prefix.  Arguments to options that specify a size threshold of some sort may be arbitrarily large decimal or hexadecimal integers

Manual page gcc(1) line 1 (press h for help or q to quit)

```

Topic 6: Process Management

```
student@DESKTOP-UIMU3UD:~$ ps
  PID TTY          TIME CMD
   92 tty1      00:00:00 bash
  108 tty1      00:00:00 ps
```

```
student@DESKTOP-UIMU3UD:~$ ps -ef
UID          PID  PPID  C STIME TTY          TIME CMD
root           1      0  0 09:19 ?            00:00:00 /init
root           5      1  0 09:19 ?            00:00:00 plan9 --control-socket 6 --log-level 4 --serve
r-fd 7 --pipe-fd 9 --socketroot
student       92     91  0 09:43 tty1      00:00:00 -bash
student      109     92 33 09:44 tty1      00:00:00 ps -ef
```

```
student@DESKTOP-UIMU3UD:~$ top
top - 09:44:28 up 24 min,  0 user,  load average: 0.52, 0.58, 0.59
Tasks:  5 total,   1 running,  4 sleeping,   0 stopped,   0 zombie
%Cpu(s): 36.1 us, 28.6 sy,  0.0 ni, 34.2 id,  0.0 wa,  1.1 hi,  0.0 si,  0.0 st
MiB Mem : 4027.8 total,  626.1 free, 3401.7 used,  224.0 buff/cache
MiB Swap: 12288.0 total, 11907.2 free,  380.8 used.  626.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	11228	1112	1056	S	0.0	0.0	0:00.15	init(Ubuntu)
5	root	20	0	11228	800	644	S	0.0	0.0	0:00.01	init
91	root	20	0	11228	576	528	S	0.0	0.0	0:00.01	SessionLeader
92	student	20	0	14108	3844	3748	S	0.0	0.1	0:00.07	bash
110	student	20	0	17452	3896	1644	R	0.0	0.1	0:00.03	top

```
student@DESKTOP-UIMU3UD:~$ sleep 100 &
[1] 111
student@DESKTOP-UIMU3UD:~$ kill 111
[1]+  Terminated                  sleep 100
student@DESKTOP-UIMU3UD:~$ sleep 100 &
[1] 112
student@DESKTOP-UIMU3UD:~$ kill -9 111
-bash: kill: (111) - No such process
student@DESKTOP-UIMU3UD:~$
```

Topic 7: Job Control

```
student@DESKTOP-UIMU3UD:~$ echo -e '#include <stdio.h>\n#include <unistd.h>\nint main()
{ while(1) { sleep(1); } return 0; }' > hello.c
student@DESKTOP-UIMU3UD:~$ gcc hello.c -o hello
student@DESKTOP-UIMU3UD:~$ ./hello &
[1] 135
student@DESKTOP-UIMU3UD:~$ jobs
[1]+  Running                      ./hello &
student@DESKTOP-UIMU3UD:~$ fg %1
./hello
^Z
[1]+  Stopped                      ./hello
student@DESKTOP-UIMU3UD:~$ bg %1
[1]+ ./hello &
student@DESKTOP-UIMU3UD:~$
```

Topic 8: File Permissions and Links

```
student@DESKTOP-UIMU3UD:~$ touch script.sh
student@DESKTOP-UIMU3UD:~$ touch file1
student@DESKTOP-UIMU3UD:~$ chmod +x script.sh
student@DESKTOP-UIMU3UD:~$ chmod 755 script.sh
student@DESKTOP-UIMU3UD:~$ ln file1 file2
ln: failed to create hard link 'file2': File exists
student@DESKTOP-UIMU3UD:~$ ln -s file1 shortcut
ln: failed to create symbolic link 'shortcut': File exists
student@DESKTOP-UIMU3UD:~$
```

Topic 9: Disk and Memory Information

```
student@DESKTOP-UIMU3UD:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
rootfs          248G   74G  174G  30% /
none            248G   74G  174G  30% /dev
none            248G   74G  174G  30% /run
none            248G   74G  174G  30% /run/lock
none            248G   74G  174G  30% /run/shm
none            248G   74G  174G  30% /run/user
tmpfs           248G   74G  174G  30% /sys/fs/cgroup
C:\             248G   74G  174G  30% /mnt/c
D:\             196G   43G  153G  22% /mnt/d
student@DESKTOP-UIMU3UD:~$ du -sh .
296K  .
student@DESKTOP-UIMU3UD:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.9Gi         3.2Gi         736Mi        17Mi        223Mi        736Mi
Swap:         12Gi         498Mi         11Gi
```

Topic 10: User and Environment Management

```
student@DESKTOP-UIMU3UD:~$ id
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),107(netdev)
```

```
student@DESKTOP-UIMU3UD:~$ env
SHELL=/bin/bash
WSL_DISTRO_NAME=Ubuntu
NAME=DESKTOP-UIMU3UD
PWD=/home/student
LOGNAME=student
HOME=/home/student
LANG=C.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:* tar=01;31:* tgz=01;31:* arc=01;31:* arj=01;31:* taz=01;31:* lha=01;31:* lz4=01;31:* lz=01;31:* lzh=01;31:* lzma=01;31:* tlz=01;31:* txz=01;31:* tzo=01;31:* t7z=01;31:* zip=01;31:* z=01;31:* dz=01;31:* gz=01;31:* lrz=01;31:* lz=01;31:* lzo=01;31:* xz=01;31:* zst=01;31:* tzs=01;31:* b22=01;31:* bz=01;31:* tbz=01;31:* tbz2=01;31:* tz=01;31:* deb=01;31:* rpm=01;31:* jar=01;31:* war=01;31:* ear=01;31:* sar=01;31:* rar=01;31:* alz=01;31:* ace=01;31:* zoo=01;31:* cpio=01;31:* 7z=01;31:* rz=01;31:* cab=01;31:* wim=01;31:* swm=01;31:* dwm=01;31:* esd=01;31:* avif=01;35:* jpg=01;35:* jpeg=01;35:* mjpg=01;35:* mjpeg=01;35:* gif=01;35:* bmp=01;35:* pbm=01;35:* pgm=01;35:* ppm=01;35:* tga=01;35:* xbm=01;35:* xpm=01;35:* tif=01;35:* tiff=01;35:* png=01;35:* svg=01;35:* svgz=01;35:* mng=01;35:* pcx=01;35:* mov=01;35:* mpg=01;35:* mpeg=01;35:* m2v=01;35:* mkv=01;35:* webm=01;35:* webp=01;35:* ogm=01;35:* mp4=01;35:* m4v=01;35:* mp4v=01;35:* vob=01;35:* qt=01;35:* nuv=01;35:* wmv=01;35:* asf=01;35:* rm=01;35:* rmvb=01;35:* flc=01;35:* avi=01;35:* fli=01;35:* flv=01;35:* gl=01;35:* dl=01;35:* xcf=01;35:* xwd=01;35:* yuv=01;35:* cgm=01;35:* emf=01;35:* ogv=01;35:* ogx=01;35:* aac=00;36:* au=00;36:* flac=00;36:* m4a=00;36:* mid=00;36:* midi=00;36:* mka=00;36:* mp3=00;36:* mpc=00;36:* ogg=00;36:* ra=00;36:* wav=00;36:* oga=00;36:* opus=00;36:* spx=00;36:* xspf=00;36:* ~00;90:* #00;90:* bak=00;90:* crdownload=00;90:* dpgk-dist=00;90:* dpgk-new=00;90:* dpgk-old=00;90:* dpgk-tmp=00;90:* old=00;90:* orig=00;90:* part=00;90:* rej=00;90:* rpmnew=00;90:* rpmorig=00;90:* rpmsave=00;90:* swp=00;90:* tmp=00;90:* ucf-dist=00;90:* ucf-new=00;90:* ucf-old=00;90:
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=student
SHLVL=1
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/napd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/mnt/c/Program Files/WindowsApps/CanonicalGroupLim
ited.Ubuntu.2404.1.68.0_x64_79rhkp1fndgsc:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPo
werShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files (x86)/dotnet:/mnt/c/Users/Abinash/Videos/Captures/New folder:/mnt/c/Users/
Abinash/Videos/Captures/New folder/Library/mingw-w64/bin:/mnt/c/Users/Abinash/Videos/Captures/New folder/Library/usr/bin:/mnt/c/Users/Abinash/Vi
deos/Captures/New folder/Library/bin:/mnt/c/Users/Abinash/Videos/Captures/New folder/Scripts:/mnt/c/Users/Abinash/AppData/Local/Microsoft/Window
sApps:/mnt/c/Users/Abinash/Pictures/BlueStacks NEW SONGS BASE CAMPUS/PyCharm Community Edition 2021.2.2/bin:/mnt/c/Users/Abinash/Pictures/BlueSt
acks NEW SONGS BASE CAMPUS/Microsoft VS Code/bin:/snap/bin
HOSTTYPE=x86_64
_=/usr/bin/env
```

```
student@DESKTOP-UIMU3UD:~$ export PATH=$PATH:.  
student@DESKTOP-UIMU3UD:~$ alias ll='ls -l'
```

```
student@DESKTOP-UIMU3UD:~$ ll  
total 216  
-rw-r--r-- 1 student student 76 Aug 8 15:19 Hello.c  
-rw----- 1 student student 155 Aug 8 15:14 Hello.c.save  
drwxr-xr-x 1 student student 4096 Aug 9 08:32 OSLab  
-rw-r--r-- 2 student student 22 Aug 9 11:39 a.txt  
-rw-r--r-- 1 student student 22 Aug 9 11:39 b.txt  
-rwxr-xr-x 1 student student 16320 Aug 11 06:00 cmd_exec  
-rw-r--r-- 1 student student 557 Aug 11 05:59 cmd_exec.c  
-rwxr-xr-x 1 student student 16008 Aug 9 06:07 exec_example  
-rw-r--r-- 1 student student 159 Aug 9 06:07 exec_example.c  
-rw-r--r-- 1 student student 0 Aug 11 09:52 file1  
-rw-r--r-- 2 student student 22 Aug 9 11:39 file2  
-rwxr-xr-x 1 student student 16184 Aug 11 09:21 file_copy  
-rw-r--r-- 1 student student 532 Aug 11 09:20 file_copy.c  
-rwxr-xr-x 1 student student 16088 Aug 9 05:23 fork_exec  
-rw-r--r-- 1 student student 259 Aug 9 05:23 fork_exec.c  
-rwxr-xr-x 1 student student 16088 Aug 9 06:10 fork_exec2  
-rw-r--r-- 1 student student 264 Aug 9 06:09 fork_exec2.c  
-rwxr-xr-x 1 student student 16008 Aug 9 05:17 fork_test  
-rw-r--r-- 1 student student 101 Aug 9 05:17 fork_test.c  
-rwxr-xr-x 1 student student 15960 Aug 11 09:50 hello  
-rw-r--r-- 1 student student 87 Aug 11 09:49 hello.c  
-rw-r--r-- 1 student student 43 Aug 11 09:20 input.txt  
drwxr-xr-x 1 student student 4096 Aug 11 04:08 myshell  
-rwxr-xr-x 1 student student 16064 Aug 8 15:26 name  
-rw-r--r-- 1 student student 154 Aug 8 15:25 name.c  
-rw----- 1 student student 77 Aug 9 11:35 nano.25.save  
-rw-r--r-- 1 student student 43 Aug 11 09:21 output.txt  
-rwxr-xr-x 1 student student 16008 Aug 9 05:21 parent_child  
-rw-r--r-- 1 student student 226 Aug 9 05:20 parent_child.c  
-rwxr-xr-x 1 student student 16104 Aug 9 06:02 parent_child_pid  
-rw-r--r-- 1 student student 287 Aug 9 06:01 parent_child_pid.c  
-rwxr-xr-x 1 student student 16000 Aug 8 15:28 pid  
-rw-r--r-- 1 student student 108 Aug 8 15:28 pid.c  
-rwxr-xr-x 1 student student 1270 Aug 9 11:39 run_all.sh  
-rwxr-xr-x 1 student student 0 Aug 11 09:52 script.sh  
lrwxrwxrwx 1 student student 5 Aug 9 11:39 shortcut -> a.txt  
-rw-r--r-- 1 student student 0 Aug 9 12:00 test.c  
-rwxr-xr-x 1 student student 16048 Aug 9 06:04 wait_example  
-rw-r--r-- 1 student student 273 Aug 9 06:04 wait_example.c  
-rwxr-xr-x 1 student student 16088 Aug 9 05:26 zombie  
-rw-r--r-- 1 student student 303 Aug 9 05:26 zombie.c
```